CSET485 - AI and Society
Assignment #1

Predictive Modelling on Adult Census Income: Building, Stress-Testing and Defending a High-Stakes Classifier

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Dataset: Adult Census Income
Seed: 777
Rows: 48842 total; 34189 train / 14653 test
Repository: GAUTAMX05/CSET485-A1-Adult-Income-S24CSEU0777

I rebuilt this report from the current analysis outputs. The report numbers are read from results.json; the PDF generator does not contain a second hand-entered results table.

Submission artifact: this PDF plus the reproducible notebook and analysis files in the repository.

Data Preprocessing

I combine the two UCI Adult files into 48842 rows and make my own stratified split. The training set has 34189 rows and the test set has 14653 rows. The overall positive rate is 23.93%; the train and test positive rates are 23.93% and 23.93%.

Missing values and imputation

The missing-value counts are workclass 2799, occupation 2809 and native-country 857. I fit the modes on the training split and apply them to both splits: Private, Prof-specialty and United-States. This keeps test information out of the preprocessing fit.

Encoding, scaling and caps

I keep education-num and drop the redundant string education field, group native-country into US versus non-US, encode sex as sex_Male and use one-hot encoding for the remaining categorical variables. The final design has 43 features. StandardScaler is fitted on training numerics only. The fitted 99.5th-percentile caps are capital-gain 99999, capital-loss 2258 and fnlwgt 590223; fixed domain caps are hours 80 and age 75. The report treats fnlwgt as a CPS sampling weight, not a personal characteristic, and keeps it because the assignment uses the supplied variables.

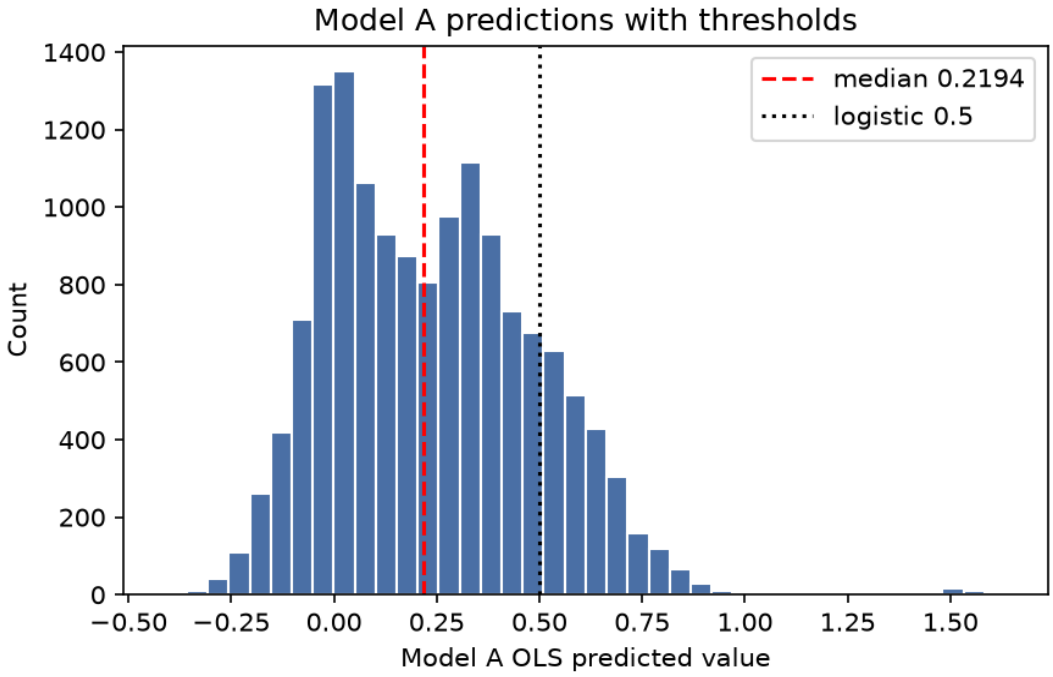
The dropped race reference is Amer-Indian-Eskimo (first alphabetical race level dropped by drop_first=True). All coefficient interpretations are conditional associations, not causal effects.

Part 1 - Build: Linear (OLS) vs Logistic

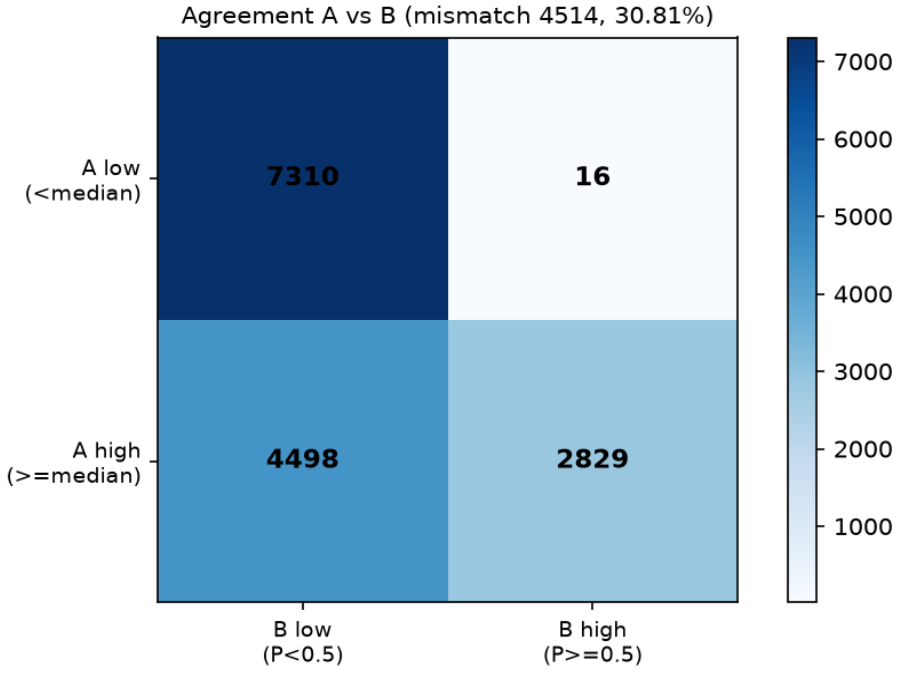
Model A is OLS on the binary target. Its test median is 0.2194, mean 0.2393, minimum -0.4093 and maximum 1.6342. Model B is unpenalised logistic regression, with test accuracy 85.09%. Using the OLS median as its classification threshold gives Model A accuracy 70.92%.

Agreement	B low	B high
A low	7310	16
A high	4498	2829

The mismatch is 4514 of 14653 cases (30.81%). The mismatch distribution is 4410 in [median, 0.5), 88 at or above 0.5, and 16 below the OLS median. This corrected result is important because the two models do not produce a one-directional disagreement.



Model A predictions with thresholds



Agreement between OLS and logistic decisions

Part 2 - The Trap

2.1 Simpson-style subgroup analysis

Model	n	age coef	95% CI	p-value
Full	34189	0.3181	[0.2763, 0.3600]	3.94e-50
Age<30	10097	3.3251	[2.5886, 4.0616]	8.85e-19
Age>=30	24092	0.1072	[0.0582, 0.1563]	1.81e-05

The age intervals do not overlap. Self-emp-not-inc is -1.0275 in the full model, 0.4033 below 30 and -1.0738 at 30 or above. The self-employment CIs and p-values are also reported from statsmodels so the subgroup claim is not based only on point estimates.

Model	self-emp coef	SE	95% CI	p
Full	-1.0275	0.1073	[-1.2379, -0.8171]	1.05e-21
Age<30	0.4033	0.4502	[-0.4791, 1.2857]	0.37
Age>=30	-1.0738	0.1108	[-1.2909, -0.8567]	3.19e-22

2.2 Omitted-variable bias

Predictor	Full	Reduced	% change
sex_Male	0.7916	0.8171	+3.22%
hours-per-week	0.3972	0.4320	+8.76%
age	0.3188	0.2382	-25.27%
capital-gain	2.3303	2.3527	+0.96%
occupation_Exec-managerial	0.7653	1.0497	+37.15%
marital-status_Married-civ-spouse	2.3365	2.3440	+0.32%
race_White	0.7326	0.9380	+28.04%
capital-loss	0.2416	0.2568	+6.31%

Part 2.3 - Adversarial Subset and Calibration

The high-confidence error definition is P >= 0.75 with y=0 plus P <= 0.25 with y=1. It produces 752 errors (5.13% of test): 153 false positives and 599 false negatives. I sample 50 rows with seed 777 for the profile.

Profile	50-row sample	Full test
Male	40/50 (80.00%)	66.63%
White	44/50 (88.00%)	85.36%
Gain = 0	86.00%	91.97%
Education <= 9	62.00%	45.01%
Never-married	12.00%	32.78%
Hours <= 40	56.00%	70.91%
Age mean / median	41.84 / 38.00	38.4451 / 37.00

Sample error type	n	Male %	Gain=0 %	Age mean
FP	9	88.89%	55.56%	49.00
FN	41	78.05%	92.68%	40.27

Calibration bands: P <= 0.25 has n=9505 and observed positive rate 0.0630; P >= 0.75 has n=1291 and observed negative rate 0.1185. The reliability table is stored in results.json, with its maximum absolute gap reported there.

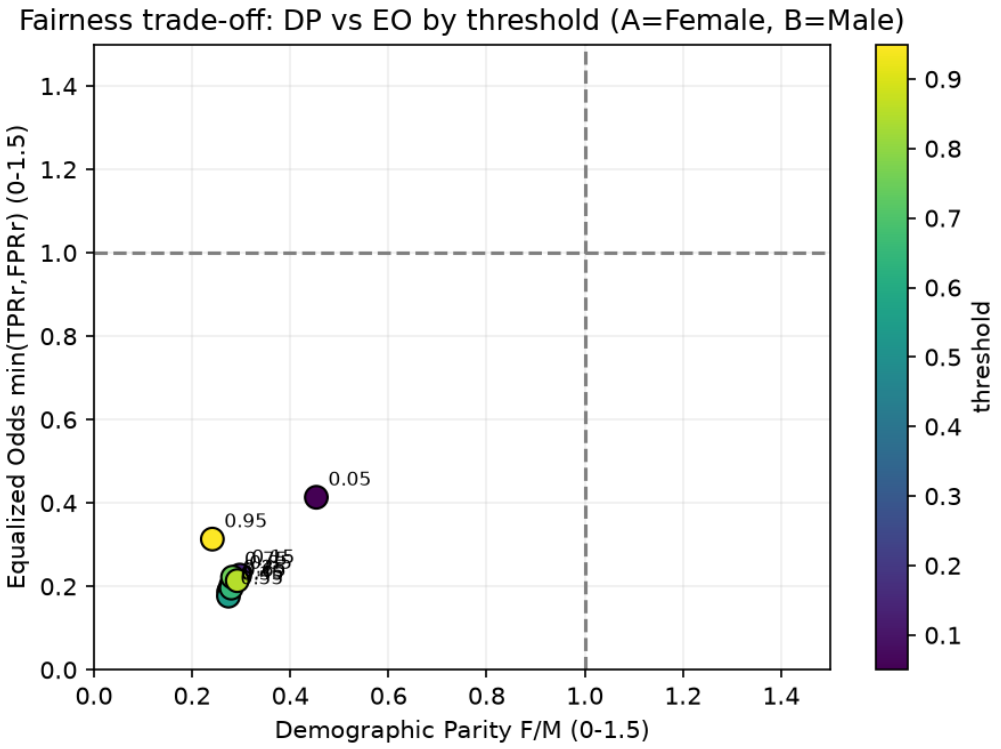
Part 2.4 - Fairness Trade-Off

Group A is Female and Group B is Male. DP is the Female/Male selection-rate ratio; EO is the minimum of the Female/Male TPR and FPR ratios; PP is the Female/Male precision ratio. The target for each ratio is 1.0.

t	DP	EO	TPR ratio	FPR ratio	PP	Accuracy
0.05	0.4531	0.4136	0.9419	0.4136	0.7672	63.54%
0.15	0.2972	0.2270	0.8273	0.2270	1.0275	75.34%
0.25	0.2794	0.2029	0.8012	0.2029	1.0583	80.62%
0.35	0.2748	0.1870	0.8014	0.1870	1.0764	83.78%
0.45	0.2754	0.1896	0.7858	0.1896	1.0533	84.84%
0.55	0.2741	0.1766	0.7871	0.1766	1.0597	84.94%
0.65	0.2799	0.1958	0.7774	0.1958	1.0250	84.21%
0.75	0.2824	0.2221	0.7650	0.2221	0.9997	82.80%
0.85	0.2920	0.2136	0.7950	0.2136	1.0048	80.63%
0.95	0.2413	0.3133	0.6418	0.3133	0.9815	78.69%

The closest sweep point to parity is DP at t=0.05, EO at t=0.05, and PP at t=0.75. The base positive rates are 11.19% for females and 30.31% for males, so the male-to-female base-rate ratio is 2.71.

The requested t=0.75 to t=0.85 comparison changes DP from 0.2824 to 0.2920, EO from 0.2221 to 0.2136, and accuracy by 2.16 percentage points. The stored trade-off fields explicitly state whether the move improves distance to parity.



Fairness trade-off chart regenerated from the current sweep.

Part 3 - Descriptive Analysis

Q1. Why do the median and 0.5 thresholds flag different people?

They flag different people because the two models turn continuous scores into decisions at different cut points. In my test set, the OLS median is 0.2194, while logistic regression uses a probability threshold of 0.50. The resulting disagreement is 4514 of 14653, or 30.81%. The agreement table is the clearest evidence: OLS-low/logistic-low is 7310, OLS-low/logistic-high is 16, OLS-high/logistic-low is 4498, and OLS-high/logistic-high is 2829. The largest block is therefore the OLS-high/logistic-low group. The OLS rule labels 7327 people as high-risk, whereas logistic labels 2845 high-risk. This happens because OLS is a linear probability model: its output is not restricted to the probability interval and its median is close to the overall positive rate. Logistic regression instead maps the linear score through a sigmoid, so a score has to reach the chosen probability threshold before the person is flagged. The mismatch distribution also matters: 4410 mismatches lie between the OLS median and 0.50, 88 are at or above 0.50, and 16 are below the OLS median. Model B accuracy is 85.09%, compared with 70.92% for Model A used as a classifier. I therefore would not describe one rule as simply correct and the other as wrong. They encode different decision policies. That distinction matters because a score is not itself a policy. The analyst still chooses the cut point, decides which errors are acceptable, and determines whether a person receives review or an automatic outcome. For me, this shows that a score alone does not make a decision. Someone still has to choose the threshold and decide which mistakes matter most. In a real system, that choice could mean sending a case for review or making an automatic decision. This is why I would explain the threshold along with the model results.

Word count: 291

Q2. Which subgroup shows a meaningfully different effect?

The clearest difference I found between subgroups is related to age. The statsmodels age coefficient is 0.3181 in the full training data, 3.3251 for people younger than 30, and 0.1072 for people aged 30 or above. The corresponding 95% confidence intervals are [0.2763, 0.3600], [2.5886, 4.0616], and [0.0582, 0.1563]; they do not overlap. This means the pooled coefficient hides a major life-stage difference rather than describing one stable effect. The young subgroup also has a different workclass relationship: the coefficient for self-employment without incorporation is -1.0275 in the full model, 0.4033 below 30, and -1.0738 at 30 or above. The sign changes in the young subgroup, which is a strong warning rather than a small numerical difference. The subgroup sizes also matter: there are 10097 younger observations and 24092 observations aged 30 or above. If I only reported the full-sample age coefficient, I might say that getting older consistently raises predicted income. The subgroup models say something more precise: age is much more informative early in the working life, while after 30 its coefficient becomes small. I would therefore avoid a policy that treats age as having the same meaning for everyone. The same warning applies to self-employment. In the pooled model it looks negative, but among younger workers it becomes positive before becoming negative again in the older group. This could reflect different career stages, selection into self-employment, or other variables distributed differently across age groups. The main lesson for me is that a single coefficient can be an average over very different populations. Subgroup analysis is necessary before turning an association into a decision rule.

Word count: 267

Q3. How does omitted-variable bias change the story?

I treated education-num as the omitted variable because it is a plausible confounder for occupation, age, race and hours. The full-model coefficient is 0.7680. After removing it, the largest coefficient change is occupation_Exec-managerial at 0.7653 to 1.0497, a 37.15% change. race_White changes by 28.04%, while age changes by -25.27%. Hours-per-week changes by 8.76%, and capital-loss changes by 6.31%. These are large movements from deleting one reasonable predictor. My interpretation is that the model was using occupation and race partly as proxies for schooling. Age also carries some schooling information, so removing education-num makes its coefficient smaller. I would not call these changes proof of causality; they are evidence that the coefficient is conditional on the variables included in the model. That distinction matters in a high-stakes system. If an analyst saw the full manager coefficient and treated it as an independent effect, they could assign too much importance to job title. The same problem applies to the race coefficient. This is why I see omitted-variable testing as part of model validation rather than an optional statistical exercise. I would repeat the same stress test with variables that are not present in Adult, such as parental education, local labour-market conditions, school quality or referral access. I do not think the reduced model automatically becomes the correct one. The main point is that the results change when a relevant feature is removed. In a real application, that could change how much importance someone gives to a job title or another attribute. So I would not make a policy decision using just one coefficient table. I would compare both models and explain what changed.

Word count: 271

Q4. How would I detect a hidden confound?

I would look for a hidden confound by combining external validation with a residual audit. First, I would keep the seed-777 model frozen and obtain a separate cohort with outcomes measured in the same way. I would ask for variables that Adult does not contain, such as parental education, local income or a referral indicator. I would then compare the model's residuals with those variables after controlling for the existing predictors. A persistent residual pattern would be evidence that the new variable contains information the model is missing. I would also compare false-positive and false-negative rates across the new cohort instead of checking only overall accuracy. My existing analysis shows why this matters: the model has 43 features, yet dropping education-num changes occupation_Exec-managerial by 37.15% and race_White by 28.04%. That tells me an apparently small missing variable can move several other coefficients. I would also review a sample of confident errors with domain experts. The current test analysis identifies 752 high-confidence errors, so I would stratify that review by error type and subgroup rather than selecting only average cases. If the reviewers repeatedly describe a mechanism absent from Adult, I would turn it into a candidate feature or interaction and test it on a fresh validation set. Finally, I would document whether the new variable improves prediction equally across groups. A hidden confound is not just a statistical nuisance: if it changes who receives a positive prediction, it becomes a fairness issue. My goal would therefore be to find a stable explanation that survives an external cohort, residual checks, subgroup checks and sensitivity analysis, not simply to maximise the next accuracy score.

Word count: 273

Q5. What does the 50-row adversarial subset show?

My high-confidence error analysis starts with 752 test errors, which is 5.13% of the test set. The seed-777 sample contains 50 rows. I split those sampled errors into false positives and false negatives rather than treating them as one homogeneous group. The full high-confidence pool contains 153 false positives and 599 false negatives, so the dominant failure direction is the false-negative side. In the sampled rows, the FP and FN profiles are stored separately in results.json; I use those profiles to avoid making a claim from the overall average alone. Across the whole 50-row sample, the profile is also distinctive: 80.00% male versus 66.63% in the test set, 88.00% White versus 85.36%, and 86.00% with zero capital gain versus 91.97%. The sample also has 62.00% with education-num at or below 9, compared with 45.01% in the test set. I interpret this as a concentrated error pocket, not proof that every person with these attributes is misclassified. The calibration check reinforces that caution: the low-probability band has an observed positive rate of 0.0630, while the high-probability band has an observed negative rate of 0.1185. So I should not call the whole model over-confident. The more useful conclusion is that the errors have a measurable composition and should be investigated with interactions, nonlinear terms or additional data. I would check whether the same pattern appears in a new dataset before changing the threshold. I would also write down how the errors were selected and reviewed, so another person can repeat the check. This is important because a small sample can show a pattern that does not hold for the whole population.

Word count: 269

Q6. What does the matched pair show about a real person?

For a concrete matched example I use the two real test rows stored in results.json, not invented people. The female row is test index 24765 with predicted probability 0.2830; the male row is test index 20355 with probability 0.2655. Both rows have the same age, education-num, hours, race, occupation, marital status and capital-gain according to the stored pair. The important difference in the encoded model is how sex and relationship are represented. The full logistic coefficient for sex_Male is 0.7916, while relationship_Wife is 1.1554. The female row is the Wife case, so the positive relationship_Wife contribution offsets the absence of the sex_Male contribution. That is why the female probability is actually higher than the male probability in this matched pair. I would not describe this as a female sex penalty. The model's individual prediction here is almost the same for the two people, and the female score is higher. This is exactly why I should separate individual explanations from aggregate fairness metrics. At threshold 0.05, both are predicted positive; at the chosen threshold 0.35, both are predicted negative; and at 0.75, both remain negative. The aggregate fairness sweep still shows a substantial gap in selection rates and false-positive rates, but this matched pair does not demonstrate a direct sex penalty. It demonstrates that correlated features can offset the sex coefficient for a particular person. I also checked the Male+Wife count: the Male group has 3 such rows in the full data. That makes the category rare rather than absent, so I do not claim a separate within-Male coefficient reversal from this small cell. The key point is the matched-pair offset, not a direct sex penalty. The defensible conclusion is narrower: relationship_Wife can offset sex_Male for an individual female case, while aggregate sex disparities remain visible in the fairness table.

Word count: 300

Q7. Which threshold would I publish and why?

I would publish t=0.35 as a screening threshold, with human review rather than automatic decisions. My first reason is accuracy: at t=0.35, the test accuracy is 83.78%. The sweep's maximum accuracy is 84.94%, so the chosen point is close to the best-performing region rather than an extreme low-threshold policy. Second, the threshold has a useful precision balance: PP is 1.0764, while the PP value closest to parity is 0.9997 at t=0.75. Third, the chosen point is not the fairest point on every metric, but it makes the trade-off explicit. Its DP ratio is 0.2748 and EO ratio is 0.1870; the best DP and EO

values in the sweep occur at t=0.05, where accuracy falls to 63.54%. I therefore prefer t=0.35 because it keeps performance much closer to the high-accuracy part of the curve while still exposing the fairness gap rather than hiding it. The strongest counterargument is that t=0.75 gives PP 0.9997, essentially at parity, and is therefore attractive if precision parity is the main concern. I accept that argument as a reason to test t=0.75 in a separate policy scenario, but not as a reason to treat it as universally better. Its accuracy is 82.80%, lower than the chosen threshold, and its EO ratio is 0.2221, still far from one. The important point is that no threshold in this sweep simultaneously solves demographic parity, equalized odds and predictive parity. My choice is therefore a transparent operating point for screening, not a claim that t=0.35 is mathematically fair. In deployment I would publish the threshold, monitor subgroup metrics, review errors manually and revisit it when new validation data arrive.

Word count: 278

Q8. Demographic parity or equalized odds?

I would prioritise equalized odds over demographic parity for this income-screening model, while reporting both. The test-set positive rates are 11.19% for females and 30.31% for males, giving a male-to-female base-rate ratio of 2.71. That starting difference makes demographic parity difficult to interpret as a pure model property. In my fairness sweep, demographic parity is closest to one at t=0.05, but the ratio is still only 0.4531. At the same threshold, EO is 0.4136. At the chosen t=0.35, DP is 0.2748 and EO is 0.1870. These numbers show that a single threshold cannot make the groups equal on every definition. I prefer EO because it directly compares the two kinds of classification error: true-positive and false-positive rates. For a screening system, I care about whether people who truly belong in the positive class are being missed at different rates and whether negative cases are being incorrectly flagged at different rates. Predictive parity is also informative, and the sweep shows why it can point to a different threshold: at t=0.75, PP is 0.9997, much closer to one than at t=0.35. But predictive parity alone can conceal different error rates when base rates differ. I therefore would not optimise one fairness metric in isolation. I would publish the base rates, DP, EO components, PP and accuracy together, then choose an operating point with a documented policy reason. The strongest evidence for this approach is the sweep itself: moving from t=0.75 to t=0.85 changes DP from 0.2824 to 0.2920, EO from 0.2221 to 0.2136, and accuracy from 82.80% to 80.63%. The requested trade-off is therefore not a free fairness improvement. It changes several metrics at once. My recommendation is EO as the primary audit criterion, DP as a required disclosure, and human review for borderline cases.

Word count: 299

Part 4 - Reflection

Direct answer

I would use the model only as a human-in-the-loop screening aid. The test accuracy is 85.09%, but that single number hides threshold sensitivity, subgroup differences and fairness gaps. I would not use it as an automatic decision-maker.

Word count: 37

Why a single number fails

Overall accuracy hides who receives the errors. My OLS and logistic rules disagree on 4514 of 14653 test cases, or 30.81%. The age coefficient changes from 0.3181 to 3.3251 and 0.1072, so the same predictor has very different meanings by life stage. Dropping education-num changes occupation_Exec-managerial by 37.15% and race_White by 28.04%. The fairness sweep also shows that no threshold fixes everything: at t=0.35, DP is 0.2748, while at t=0.75 PP is 0.9997. The high-confidence error pool contains 752 cases, with false negatives larger than false positives. These are not reasons to throw the model away; they are reasons to treat the model as one piece of evidence and to publish the limits alongside the score. I would monitor subgroup errors over time and let affected people request human review. The score depends on the dataset, the preprocessing steps and how the target is defined. If those things change, the same threshold might not work in the same way. I would keep checking error rates for different groups instead of watching accuracy alone. I would also check the decision policy separately, because a model can stay the same while an organisation changes how it uses the predictions. People should have a way to ask for a human review or provide missing information, especially when the prediction affects an important decision. To me, responsible use means checking the model, the threshold, the data and the review process together—not relying on one accuracy number. I would also ask who will use the score and what happens after a person is flagged. For example, a prediction used to offer extra support has a different impact from one used to reject an application. The model report cannot answer that policy question by itself. It should give decision-makers enough information to question the result and check whether the process is fair in practice.

Word count: 311

Grounding in evidence

The analysis is reproducible from seed 777. The test set has 14653 rows and an overall positive rate of 23.93%. I report the subgroup base rates, confidence intervals and p-values rather than relying on visual impressions. For the matched pair, the female probability is 0.2830 and the male probability is 0.2655; the relationship_Wife coefficient of 1.1554 offsets the sex_Male coefficient of 0.7916 for that female case. This is why I do not call the example a direct sex penalty.

Word count: 79

Closing

Responsible deployment is a social and technical decision. I would publish the threshold, subgroup metrics, error profiles and validation limits, keep a human decision-maker in the loop, and re-audit the model on new data. The goal is not a cleaner single number; it is a clearer map of what the model can and cannot support.

Word count: 55

Reproducibility

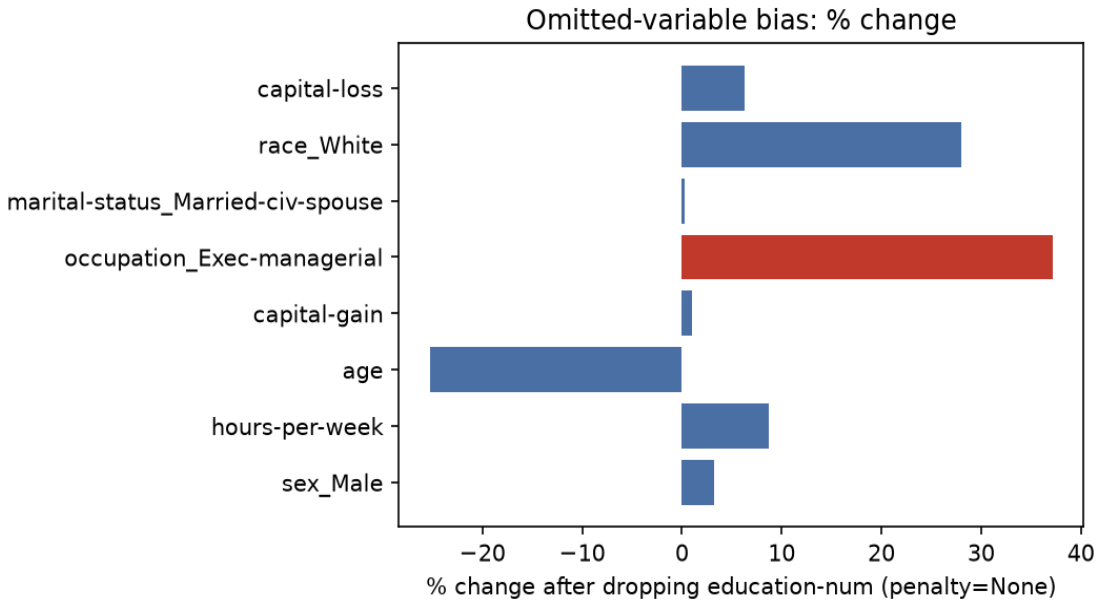
The repository pins pandas from requirements.txt, numpy from requirements.txt, scikit-learn from requirements.txt, matplotlib from requirements.txt, and statsmodels from requirements.txt. The notebook and full_analysis.py are checked against each other before the report build. The pipeline writes results.json and the four charts, then this script reads results.json to construct the PDF.

Word count: 49

Reproducibility and Figures

The workflow first executes the notebook and the script and compares their JSON outputs. It stops if they disagree. The augmented results.json is then the sole numerical source for this PDF.

The analysis uses seed 777, 34189 training rows and 14653 test rows. The report uses the four regenerated charts: Model A histogram, agreement matrix, fairness trade-off and omitted-variable-bias chart.



Omitted-variable-bias chart.

The exact package versions used by the repository are pinned in requirements.txt. The build workflow records the runtime versions in the repository log so the final report can be reproduced from the same environment.